

Pranav Arora

pranav2vis@gmail.com · +65 8515 7891 · linkedin.com/in/pranavarora63 · pranavarora.vercel.app

Profile

Applied AI Scientist building governed, evaluation-first AI systems for government and enterprise — agentic workflows, retrieval and release safety, taken from research through production deployment. Six years across applied AI, semiconductor manufacturing and enterprise ML in Singapore and Abu Dhabi, with delivered impact including \$10M annual revenue from RL scheduling and platforms serving 5,000+ users.

Experience

Applied AI Scientist

Jul 2026 – Present

Inception, a G42 company · Abu Dhabi, UAE

Machine-Readable Government (MRG) — Policy Intelligence & Compliance

- Building a multi-tenant policy-compliance platform that transforms uploaded government policies into machine-readable artifacts and applies them to evidence-backed, reviewable case workflows.
- Engineering bilingual PDF-to-markdown and policy-graph processing with governed extraction review, versioned provenance, explainable corpus-health checks, and controlled publication.
- Developing agentic case-processing workflows with evidence retrieval, compliance assessment, human review, configuration pinning, OpenAI Agents SDK sandbox execution, and isolated Docker/AKS worker jobs.

Governed Configuration, Evaluation & Release Safety

- Building Configuration Studio lifecycles for customer-scoped use cases with versioned drafts, deterministic prompt-safety assessment, approval workflows, audit history, and separate runtime activation.
- Developing an Eval Workbench with approved ground-truth suites, repeated-run stability measurement, stage-level quality analysis, exact-version release gates, and paired regression comparisons.

Enterprise AI Platform Engineering

- Delivering the platform with FastAPI, PostgreSQL, React/TypeScript, Microsoft Entra ID, row-level security, Azure/AKS infrastructure, immutable run fencing, and privacy-safe production observability.

Scope: multi-tenant policy platform across bilingual (Arabic/English) corpora · governed extraction, evaluation and release-gate lifecycle · agentic case processing on isolated Docker/AKS workers

Senior ML Engineer — AI Engineering Team

Aug 2024 – Jul 2026

Hewlett Packard Enterprise · Singapore

Text-to-SQL AI Platform

- Architected a natural-language query platform for 200+ non-technical users across 75+ tables using LangChain, GPT-4o, schema-aware generation, Redis caching, and database access controls.
- Deployed FastAPI services on HPE Ezmeral Kubernetes with Keycloak SSO, row-level security, Azure AI Foundry model deployments, and team-scoped RBAC.

Impact: 85% query accuracy · 2,000+ queries/week · 7 business units · query time reduced from 10 minutes to 1 minute

K8s Watcher — Agentic Monitoring & Remediation

- Designed and deployed an agentic AI system that monitored live Kubernetes state, diagnosed incidents with GPT-4o, and generated context-aware remediation guidance.

Impact: 50+ incidents diagnosed/week · mean time to resolution reduced by 70%

Document Planning Hub & OneAI Platform

- Built a LangGraph multi-agent validation service for PDF, PPTX, Excel, and DOCX documents, using specialist agents and an aggregator to produce structured quality reports.
- Led shared AI engineering standards across eight teams, including CI/CD templates, MLOps patterns, Docker conventions, review practices, and an Azure OpenAI on-call assistant.

Impact: 5,000+ users · document errors reduced by 80% · deployment failures reduced by 60% · time-to-production reduced from 6 weeks to 2 weeks

Real-Time Address Correction

- Deployed an Azure-hosted LLM service integrating Google Maps API and Azure OpenAI for high-throughput, sub-second delivery-address correction.

Impact: supply-chain fallout reduced by 35% · manual interventions reduced by 50%

Pranav Arora

Applied AI Scientist · Abu Dhabi, UAE

Experience

Data Scientist — Smart Manufacturing & AI Team

Jan 2022 – Aug 2024

Micron Technology · Singapore

- Optimised semiconductor wafer scheduling using PPO reinforcement-learning agents with Ray RLlib across a 70-machine distributed training cluster, owning design through production deployment.
- Deployed a recommendation pipeline updating the production UI every two hours and refactored NumPy simulation code to reduce runtime by 60%.
- Fine-tuned an LLM chatbot on 10K+ internal manufacturing documents, achieving BLEU 0.82 and 80% first-contact resolution.
- Standardised planning and operations master data, reducing maintenance by 80% and time spent finding metrics by 50%.

Impact: 0.5% wafer-production increase → \$10M annual revenue · queueing time reduced by 15%

Data Scientist — Data Analytics Team

Aug 2020 – Jan 2022

Dentsu International · Singapore

- Developed an 85%-accuracy customer purchase-propensity model and integrated it into live marketing campaigns.
- Deployed a data catalogue to Azure AKS with Terraform, ingesting 10,000+ Azure Synapse datasets for enterprise governance.
- Built ROAS prediction models and dashboards that reduced post-campaign analysis time by 50% and media-planning costs by 20%.

Education

Master of IT in Business — Major: Artificial Intelligence, Singapore Management University

Jan 2019 – May 2020

Bachelor of Computer Science — University of Petroleum and Energy Studies, India

Aug 2014 – Apr 2018

Skills

Applied AI & Agents: OpenAI Agents SDK, LangGraph, LangChain, multi-agent orchestration, RAG, GPT-4o, prompt engineering, LangSmith

Evaluation & Governance: LLM-as-judge, RAGAS, release regression, prompt-safety controls, human-in-the-loop review, auditability, MLflow

ML & Modeling: PPO, Ray RLlib, QLoRA/PEFT, PyTorch, TensorFlow, Hugging Face, XGBoost, SHAP/LIME

Cloud & Infrastructure: Azure AI Foundry, Azure OpenAI, AKS, Kubernetes, Docker, Terraform, AWS, GCP, Keycloak, Microsoft Entra ID

MLOps & Backend: FastAPI, PostgreSQL, row-level security, KServe, Prometheus, Grafana, GitHub Actions, Azure DevOps CI/CD

Data & Interfaces: PostgreSQL/pgvector, Qdrant, Redis, Snowflake, Airflow, SQL, Python, React, TypeScript, REST APIs

Certifications & Publications

Harvard Business School Online — Business Analytics

Innovations in Software Defined Networking and Network Virtualizations — Publication